2024-Winter Paper 2 – Urology

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Q1. Pathophysiology, classification and management of LUTS.

Type: LAQ (25 marks) (medical and surgical management of BPH also answered in 2015-Summer Paper 2, Q1)

Definition

- **LUTS (lower urinary tract symptoms):** symptoms of the storage, voiding and post-micturition phases; not disease-specific; up to half of older men.
- **BPH (benign prostatic hyperplasia)** is histology; BPE is enlargement; BPO is prostatic obstruction; **BOO (bladder outlet obstruction)** has any cause. Not every man with LUTS has BPO.

Classification

1. By phase (ICS, International Continence Society)

Storage	Voiding	Post-micturition
Frequency, urgency, **nocturia** (waking ≥1 time to void), urgency incontinence	Weak or split stream, hesitancy, intermittency, straining, terminal dribble	Feeling of incomplete emptying, post-micturition dribble

- Storage symptoms form **OAB (overactive bladder)**: urgency ± urgency incontinence, usually with frequency and nocturia; most bothersome.

2. Functional (Wein)

	Because of the bladder	Because of the outlet
Failure to store	DO (detrusor overactivity), low compliance, hypersensitivity	Sphincter weakness (SUI, stress urinary incontinence)
Failure to empty	DUA (detrusor underactivity)	BPO, stricture, bladder neck contracture, sphincter dyssynergia

3. Severity: IPSS (International Prostate Symptom Score), 7 questions scored 0–5 plus a quality-of-life question: mild 0–7, moderate 8–19, severe 20–35.

4. By cause: prostate, bladder, urethra, neurological, polyuria, infection, stone, tumour.

Pathophysiology

- **Prostate:** BPH is an increase in cell number of stroma and glands in the **transition zone** and periurethral region. **Type 2 5α-reductase** in stromal fibroblasts makes **DHT**

(dihydrotestosterone); androgens are permissive, not mitogenic. Ageing, inflammation and metabolic syndrome contribute.

- **Static component:** gland bulk; the **capsule** transmits expansion to the urethra (TUIP, transurethral incision of the prostate, relieves obstruction without removing tissue).
- **Dynamic component: α1A-adrenergic** smooth-muscle tone; the basis of α-blockers.
- Size does not correlate with symptoms or obstruction.
- **Bladder response:** hypertrophy → compensation → decompensation. Collagen causes trabeculation and low compliance; DO causes storage symptoms; late detrusor failure causes retention. One-third keep storage symptoms after surgery.
- **Also know:** nocturnal polyuria (night output >33% of 24 h over age 65); BPO complications: AUR (acute urinary retention), UTI (urinary tract infection), stones, haematuria, renal failure.

Evaluation (EAU, European Association of Urology, men ≥40)

- **Basic:** history (comorbidity, drugs, sexual function), IPSS, DRE (digital rectal examination) and neurological check, **urinalysis (mandatory)**, PVR (post-void residual), PSA (prostate-specific antigen) if life expectancy >10 years.
- **Frequency–volume chart, 3 days:** for storage symptoms or nocturia.
- **Before treatment:** uroflowmetry (Qmax, maximum flow rate, <10 mL/s suggests BOO); ultrasound for volume and median lobe.
- **Selected:** creatinine and renal ultrasound if large PVR; cystoscopy for haematuria, stricture or before PUL (prostatic urethral lift).
- **Pressure–flow study** (separates BOO from DUA) before surgery if voided volume <150 mL, PVR >300 mL, age <50 or >80, or failed surgery.

Management: EAU algorithm

```
Bothersome LUTS?  No  -> watchful waiting, lifestyle advice
Absolute surgical indication? Yes -> surgery
Otherwise:
  Nocturnal polyuria      -> lifestyle +/- desmopressin
  Storage predominant     -> antimuscarinic or beta3-agonist
  Prostate >40 mL         -> 5ARI (5-alpha-reductase inhibitor)
                             +/- alpha1-blocker
  Others                  -> alpha1-blocker or PDE5 inhibitor
  Residual storage        -> add antimuscarinic or beta3-agonist
Failure or complication   -> surgery
```

- **Watchful waiting:** education, fluid 1.5–2 L/day (none 2 h before bed), less caffeine and alcohol, double voiding, bladder training; yearly review.

Drug	Example	Mechanism and notes
α1-blocker	Tamsulosin 0.4 mg, silodosin 8 mg, alfuzosin 10 mg	Relax prostatic smooth muscle; acts in days; dizziness, ejaculatory dysfunction, IFIS (intraoperative floppy iris syndrome)
5ARI	Finasteride 5 mg, dutasteride 0.5 mg	Lower DHT; volume falls about 20% over 6–12 months; **PSA halves**; reduces AUR and surgery
Antimuscarinic	Solifenacin, tolterodine	Block detrusor M3 receptors; for storage symptoms if PVR <200 mL
β3-agonist	Mirabegron 50 mg, vibegron	Relax detrusor during storage

Drug	Example	Mechanism and notes
PDE5I (phosphodiesterase-5 inhibitor)	Tadalafil 5 mg daily	Smooth-muscle relaxation; useful with ED (erectile dysfunction)

- **MTOPS:** progression cut by doxazosin 39%, finasteride 34%, combination 66%; **CombAT** confirmed. Combine when volume >30–40 mL and PSA >1.5 ng/mL.

Surgery

- **Indications:** refractory retention, recurrent UTI, bladder stones, recurrent haematuria, BPO-related renal insufficiency; failed medical therapy.

Prostate volume	Options
<30 mL	TUIP, TURP (transurethral resection of the prostate)
30–80 mL	**TURP (reference standard)**, bipolar TURP, PVP (photoselective vaporisation of the prostate), enucleation, PUL, Rezum (water vapour), Aquablation
>80 mL	**HoLEP (holmium laser enucleation)** or thulium enucleation, open or robotic simple prostatectomy

- Ejaculation preserved: PUL, Rezum. Anticoagulated: PVP, HoLEP. PAE (prostatic artery embolisation) in selected men.
- **TURP complications:** bleeding, **TUR syndrome** (hyponatraemia with glycine; avoided by bipolar saline), retrograde ejaculation, stricture, incontinence.
- **Also know:** iTind; female LUTS follow OAB and SUI guidelines.

Diagram to draw: BPH → BPE → BPO → BOO → bladder changes → LUTS.

Key points to remember:

- ICS: storage, voiding, post-micturition; IPSS 0–7, 8–19, 20–35.
- Static (bulk, capsule) plus dynamic (α1A) obstruction; size ≠ symptoms.
- 5ARI for >40 mL; MTOPS combination cut progression 66%.
- TURP 30–80 mL; HoLEP or simple prostatectomy for >80 mL.

Q2. Etiology, pathogenesis and management of acute renal failure.

Type: LAQ (25 marks) (acute tubular necrosis also answered in 2022-Summer Paper 2, Q3c)

Acute renal failure is now called AKI (acute kidney injury).

Definition and staging: KDIGO (Kidney Disease: Improving Global Outcomes)

- AKI = rise in serum creatinine **or** fall in urine output; either alone is enough.

Stage	Serum creatinine	Urine output
1	Rise ≥0.3 mg/dL in 48 h, or ×1.5 baseline in 7 days	<0.5 mL/kg/h for 6 h
2	×2 baseline	<0.5 mL/kg/h for 12 h
3	×3, or ≥4 mg/dL, or RRT (renal replacement therapy)	<0.3 mL/kg/h for 24 h, or anuria 12 h

- One-fifth to one-third of inpatients; mortality about 25%, about 50% if RRT is needed.

Etiology

Type	Causes
Prerenal	Hypovolaemia (bleeding, vomiting, burns), heart failure, sepsis, hepatorenal syndrome; worsened by NSAIDs (non-steroidal anti-inflammatory drugs) and ACE (angiotensin-converting enzyme) inhibitors
Intrinsic	**ATN (acute tubular necrosis):** ischaemia, contrast, aminoglycosides, cisplatin, myoglobin. **AIN (acute interstitial nephritis):** β-lactams, NSAIDs, PPIs (proton pump inhibitors), checkpoint inhibitors. Glomerulonephritis; vascular (renal artery or vein thrombosis, atheroemboli)
Postrenal	BOO (bladder outlet obstruction), bilateral ureteric obstruction or obstructed single kidney: stones, tumours, retroperitoneal fibrosis, ureteric injury at surgery

- Prerenal AKI and ATN cause about three-quarters of hospital AKI.

Pathogenesis

- **Autoregulation** holds GFR (glomerular filtration rate) by afferent dilatation (prostaglandins, myogenic reflex) and efferent constriction (angiotensin II, tubuloglomerular feedback).
- **Prerenal:** low perfusion lowers glomerular pressure. Tubules are intact, so they **conserve sodium and water**. NSAIDs block afferent dilatation; ACE inhibitors block efferent constriction. Prolonged prerenal state becomes ATN.
- **Ischaemic ATN:**
  - The medulla works near hypoxia (oxygen tension 10–20 mmHg); **S3 proximal tubule** and **medullary thick ascending limb** are most vulnerable.
  - Na^+/K^+-ATPase loses its basolateral position → sodium reabsorption fails → high sodium at macula densa → afferent constriction → GFR falls (protective).
  - Cells die (mostly apoptosis) and slough → **muddy-brown casts** block tubules; filtrate leaks back; endothelial injury and inflammation worsen hypoxia.
  - Phases: initiation → maintenance (oliguria) → recovery (polyuria).
- **Toxic ATN:** vasoconstriction plus direct tubular injury. **Contrast:** risk with age, diabetes, CKD (chronic kidney disease). **Rhabdomyolysis** (long lateral positioning): myoglobin causes toxicity, cast obstruction and hypoperfusion.
- **Postrenal:** raised tubular pressure, then afferent vasoconstriction; recovery depends on duration (dog studies of complete obstruction: full at 1 week, 70% at 2 weeks, none at 6

weeks).

Evaluation

- History (fluid loss, hypotension, drugs, contrast), volume status, bladder scan.
- **Ultrasound early** in unexplained AKI to exclude obstruction.

Index	Prerenal	ATN
Urine sodium	<20 mmol/L	>20 mmol/L
FeNa (fractional excretion of sodium)	<1%	>1%
FeUrea (on diuretics)	<35%	>35%
Specific gravity	>1.015	<1.010
Sediment	Bland	Granular casts, tubular cells
Fluid challenge	Quick recovery	Lag of days to weeks

- Creatine kinase; white cells (AIN); red cell casts (glomerulonephritis); biopsy only if recovery lags.

Management

General

- Treat the cause; **avoid hypotension** (MAP, mean arterial pressure, >65 mmHg or near baseline) and nephrotoxins; stop NSAIDs and ACE inhibitors or ARBs (angiotensin receptor blockers); adjust drug doses.
- Isotonic crystalloid for hypovolaemia; aggressive fluids in rhabdomyolysis; intravenous fluid before contrast.
- **Postrenal:** urethral or suprapubic catheter; **double-J stent or PCN (percutaneous nephrostomy)**; infected obstructed kidney is an emergency.
- AIN: stop drug ± steroids; glomerulonephritis: nephrology and biopsy.

Complications

- **Hyperkalaemia:** calcium gluconate 10 mL of 10% (protects heart) → insulin 10 U with 50 mL 50% dextrose, salbutamol nebuliser (shift) → binders, diuretics or dialysis (remove).
- **Volume overload:** loop diuretics in escalating doses. **Diuretics do not treat ATN.**
- Acidosis: bicarbonate. Uraemic bleeding: desmopressin 0.3 µg/kg.

Dialysis indications (AEIOU), refractory to medical therapy

	Indication
A	Acidosis
E	Electrolytes, mainly hyperkalaemia
I	Intoxications (lithium, salicylate, alcohols)
O	Overload with respiratory distress
U	Uraemia: encephalopathy, pericarditis, bleeding

- Intermittent haemodialysis via 10–12 Fr jugular or femoral catheter; **CRRT (continuous RRT)** if haemodynamically unstable. **STARRT-AKI (2020):** early start gave no survival benefit.

Post-obstructive diuresis

- Causes: retained urea (osmotic diuresis), volume expansion with ANP (atrial natriuretic peptide), and down-regulated aquaporin-2 and sodium transport.
- Up to 300 mL/h for about 2 days. Free oral fluids; if unable to drink, replace about 75% of output with 0.45% saline; electrolytes every 4–6 hours. Gradual bladder decompression has

no role.

Outcome

- Full or partial recovery, or dialysis dependence; persistence ≥90 days = CKD.
- **Also know:** biomarkers NGAL (neutrophil gelatinase-associated lipocalin), KIM-1 (kidney injury molecule-1); AKD (acute kidney disease, under 3 months); gadolinium (groups I and III) avoided in AKI.

Diagram to draw: tubular cell losing polarity → sloughing → casts and back-leak.

Key points to remember:

- KDIGO: creatinine ≥0.3 mg/dL or ×1.5; urine <0.5 mL/kg/h for 6 h.
- FeNa <1% prerenal, >1% ATN; ultrasound to exclude obstruction.
- Diuretics only for overload; dialysis for refractory AEIOU.
- Post-obstructive diuresis: oral fluids; replace 75% with 0.45% saline if needed.

Q3a. ALPP

Type: SAQ (10 marks) (also answered in 2020-Summer Paper 1, Q3e and 2019-Summer Paper 1, Q3b)

Definition

- **ALPP (abdominal leak point pressure):** the **intravesical pressure** at which urine leaks because of raised abdominal pressure, **without a detrusor contraction**.
- It measures sphincter strength: **lower ALPP = weaker sphincter**. Seen only in SUI (stress urinary incontinence); there is **no normal value**.
- Provoked by Valsalva (**VLPP**, Valsalva leak point pressure) or cough (**CLPP**, cough leak point pressure). VLPP is lower than CLPP, so state the method.

Technique

1. Urethral and rectal catheters; standing; fill to **150 mL**.
2. Graded Valsalva or cough.
3. Record **total Pves (vesical pressure) at visible leak**, including baseline.
4. No leak: repeat every 50 mL to capacity; then remove the urethral catheter and read pressure from the rectal line.
5. Report volume, catheter size and provocation.

Interpretation (McGuire 1993)

ALPP (cmH_2O)	Inference
<60	**ISD (intrinsic sphincter deficiency)**
60–90	Equivocal
≥90	**Urethral hypermobility**, little ISD

Clinical uses

- Separates ISD from hypermobility; low ALPP traditionally favours a sling or bulking agent over colposuspension.
- Post-prostatectomy incontinence: many men leak only once the urethral catheter is removed.

Limitations

- Falls with larger catheters and volumes; invalid with DO (detrusor overactivity); prolapse masks leak.
- ISD and hypermobility coexist; ALPP alone predicts surgical success poorly.
- **VALUE trial (2012):** office evaluation was non-inferior to urodynamics before surgery for uncomplicated SUI.

ALPP versus DLPP

ALPP	DLPP (detrusor leak point pressure)
Sphincter against abdominal pressure	Bladder pressure against outlet resistance
Low = bad	**High = bad**: >40 cmH_2O → hydronephrosis or reflux in 85% (McGuire 1981)
SUI	Neurogenic, low-compliance bladder

- **Also know:** newer term "urodynamic stress test" (ICS, International Continence Society); EAU (European Association of Urology): do not use leak point pressure to grade severity.

Diagram to draw: Pves, Pabd and Pdet traces with leak on Valsalva, flat Pdet.

> **Key points to remember:**
> - Pves at leak with strain, no detrusor contraction; start at 150 mL.
> - <60 ISD; 60–90 equivocal; ≥90 hypermobility.
> - VLPP < CLPP; catheter size and volume alter it.

Q3b. Tivozanib

Type: SAQ (10 marks)

Introduction

- Oral **VEGFR TKI (vascular endothelial growth factor receptor tyrosine kinase inhibitor)** used in advanced RCC (renal cell carcinoma).
- Targets **VEGFR-1, -2 and -3** (also c-KIT and PDGFR-β, platelet-derived growth factor receptor-β); highly potent and selective.

Mechanism and rationale

- In clear cell RCC, **VHL (von Hippel–Lindau) loss** lets HIF (hypoxia-inducible factor) accumulate, driving VEGF and angiogenesis.
- Tivozanib blocks VEGFR signalling, starving the tumour of new vessels.
- High selectivity means fewer off-target effects (less hand-foot skin reaction and diarrhoea than sorafenib).

Dose and pharmacology

- **1.34 mg orally once daily, 21 days of a 28-day cycle.**
- Long half-life (about 4–5 days); dose reduction for toxicity or hepatic impairment.

Key trials

Trial	Setting	Result
TIVO-1 (phase III)	0–1 prior therapy, versus sorafenib	PFS (progression-free survival) 11.9 vs 9.1 months; response 33% vs 23%; OS (overall survival) not better (28.8 vs 29.3 months)
TIVO-3 (phase III)	After 2–3 prior systemic therapies including a VEGFR TKI, versus sorafenib	PFS 5.6 vs 3.9 months (HR, hazard ratio, 0.73); response 18% vs 8%; OS not different (HR 0.97); benefit also after prior checkpoint inhibitor
TiNivo-2 (phase III)	After checkpoint inhibitor; tivozanib ± nivolumab	**Negative:** adding nivolumab did not improve PFS (no immunotherapy re-challenge)

Approval and place

- FDA (Food and Drug Administration) declined in 2013 because of the OS concern in TIVO-1; EMA (European Medicines Agency) approved first-line use in 2017.
- **FDA March 2021:** relapsed or refractory advanced RCC **after ≥2 prior systemic therapies**.
- NCCN (National Comprehensive Cancer Network): later-line option, mainly after immunotherapy combinations.

Adverse effects

- **Hypertension** (commonest; control before starting), dysphonia, fatigue, diarrhoea, proteinuria, hypothyroidism.
- Bleeding, arterial thrombosis, cardiac failure, **impaired wound healing** (stop before surgery), RPLS (reversible posterior leukoencephalopathy syndrome).
- **Monitoring:** blood pressure, thyroid tests, urine protein, liver tests.

Diagram to draw: VHL loss → HIF → VEGF → VEGFR, blocked by tivozanib.

Key points to remember:

- Selective VEGFR-1–3 TKI; 1.34 mg daily, 3 weeks on, 1 off.
- TIVO-3: PFS 5.6 vs 3.9 months; OS not different.
- FDA 2021: after ≥2 prior systemic therapies.
- TiNivo-2 negative; main toxicity hypertension.

Q3c. 46 XX Male

Type: SAQ (10 marks) (also answered in 2022-Summer Paper 1, Q3e)

Assumed meaning: 46,XX testicular DSD (difference of sex development), "XX male" or de la Chapelle syndrome.

Definition

- **Testes and male genitalia in a 46,XX individual.** Old term "XX sex reversal"; renamed **46,XX testicular DSD** (Chicago consensus 2006). About **1 in 20,000** males.
- In the DSD classification it is a 46,XX DSD of gonadal development, with ovotesticular DSD; **CAH (congenital adrenal hyperplasia)** is the androgen-excess type.

Genetics

Type	Mechanism	Phenotype
SRY-positive (about 80–90%)	**SRY** (sex-determining region Y) crosses from Yp to X during paternal meiosis	Usually normal male genitalia
SRY-negative	Testis genes switched on without SRY: **SOX9** duplication, SOX3, loss of RSPO1 or WNT4	More hypospadias, undescended testes, ovotestis

- SRY switches on SOX9 → Sertoli cells → **AMH (anti-Müllerian hormone)**: no uterus. Leydig testosterone virilises.
- The **AZF (azoospermia factor) region of Yq is absent**, so spermatogenesis always fails.

Clinical features

- Male identity; found at **infertility work-up**, or prenatally (cell-free DNA 46,XX, male ultrasound).
- Small firm testes, **azoospermia**, hypergonadotropic hypogonadism, gynaecomastia; progressive testicular regression.

Feature	46,XX male	Klinefelter (47,XXY)
Frequency	1 in 20,000	1 in 500–1000
Height	Normal or short	Often tall
Genital anomalies	More common	Rare
Sperm retrieval	**Never (no AZF)**	Possible by micro-TESE (microdissection testicular sperm extraction)

Evaluation

- Semen analysis; FSH (follicle-stimulating hormone) and LH (luteinising hormone) high, testosterone low or low-normal.
- **Karyotype** 46,XX; **FISH (fluorescence in situ hybridisation) for SRY**; gene panel if SRY-negative; scrotal ultrasound.
- In an infant with atypical genitalia, exclude CAH (17-hydroxyprogesterone) and ovotesticular DSD.

Management

- **Testosterone replacement** when hypogonadal (libido, bone, muscle).
- **Infertility:** sperm retrieval is futile; counsel **donor sperm or adoption**.

- Repair hypospadias or undescended testis; reduction mammoplasty for persistent gynaecomastia.
- Genetic counselling, psychological support, bone density follow-up.

Diagram to draw: X and Y chromosomes showing Xp–Yp exchange carrying SRY onto X.

> **Key points to remember:**
> - 46,XX testicular DSD: SRY on X in about 80–90%; 1 in 20,000.
> - Azoospermia always (AZF absent): no sperm retrieval.
> - Shorter than Klinefelter; FISH for SRY confirms.

Q3d. Brain stem Death: Biology and clinical implications.

Type: SAQ (10 marks) (also answered in 2022-Summer Paper 4, Q3b and 2015-Summer Paper 1, Q3d)

Definition

- **Irreversible loss of capacity for consciousness and breathing**; legal death in India (THOA, Transplantation of Human Organs Act, 1994; Rules 2014).

Biology

```
Brain injury -> oedema -> raised ICP (intracranial pressure)
 -> falling cerebral perfusion -> herniation ("coning")
 -> Cushing reflex (hypertension, bradycardia)
 -> catecholamine storm (hypertension, tachyarrhythmia)
 -> loss of sympathetic tone -> vasodilatation, hypotension
```

System	Consequence
Heart and vessels	Storm causes myocardial injury and arrhythmia, then **hypotension**
Lungs	Neurogenic pulmonary oedema
Pituitary	**Diabetes insipidus (DI)** (no ADH, antidiuretic hormone): polyuria, hypernatraemia; low thyroid hormone and cortisol
Hypothalamus	**Hypothermia** (poikilothermia)
Blood	DIC (disseminated intravascular coagulation) from brain thromboplastin
Whole body	Cytokine injury to organs; hyperglycaemia

Diagnosis

- **Preconditions:** apnoeic coma of known irreversible cause; exclude drugs, hypothermia, metabolic causes, shock.

Brainstem reflex (all absent)	Method	Cranial nerves
Pupillary	Bright light: pupils fixed	II, III
Corneal	Touch cornea: no blink	V, VII
Oculocephalic	Turn head: eyes move with head	VIII, III, VI
Oculovestibular	50 mL ice-cold water in each ear: no eye movement	VIII, III, VI
Motor in cranial nerve area	Supraorbital pressure: no grimace	V, VII
Gag and cough	Pharynx touch, tracheal suction: no response	IX, X

- **Apnoea test (last):** pre-oxygenate, disconnect ventilator with oxygen insufflation; **no breathing at $PaCO_2$ (arterial carbon dioxide tension) ≥60 mmHg**.
- **India:** 4-member Board (no transplant-team member); **two tests ≥6 hours apart**; Form 10. Ancillary tests (angiography, perfusion scan) if examination impossible.

Clinical implications: donor management

- **Rule of 100s:** systolic pressure >100 mmHg, urine >100 mL/h, PaO_2 (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L.
- **Vasopressin** (pressor, treats DI); thyroid hormone, methylprednisolone, insulin; warming; protective ventilation.

- **DBD (donation after brain death)** keeps circulation, so warm ischaemia is minimal; mandatory request for donation.

Diagram to draw: herniation sequence leading to storm, DI, hypothermia, DIC.

> **Key points to remember:**
> - Raised ICP → Cushing reflex → storm → hypotension, DI, hypothermia, DIC.
> - Six reflexes absent; apnoea at $PaCO_2$ ≥60 mmHg.
> - Two tests ≥6 h apart; Rule of 100s.

Q3e. Immunotherapy therapy for metastatic RCC

Type: SAQ (10 marks) (overall management of metastatic RCC: see 2024-Winter Paper 3, Q3b)

Rationale

- RCC (renal cell carcinoma) is immunogenic (spontaneous regressions, T-cell infiltration) and chemoresistant.
- **Checkpoints:** PD-1 (programmed death-1) on CD8 T cells binds tumour PD-L1 and switches the T cell off; CTLA-4 (cytotoxic T-lymphocyte antigen-4) on regulatory T cells damps priming. Blocking them **releases the brake** on existing anti-tumour immunity.
- VEGF (vascular endothelial growth factor) is immunosuppressive, so TKIs (tyrosine kinase inhibitors) synergise.

Cytokine era

- Interferon-α: response 10–15%. **High-dose IL-2 (interleukin-2):** durable complete response 7–9%, but capillary leak needs intensive care.

Checkpoint inhibitors: trials (all clear cell)

Trial	Regimen	Result
CheckMate 025	Nivolumab vs everolimus, after TKI	OS (overall survival) 25.0 vs 19.6 months
CheckMate 214	Nivolumab + ipilimumab vs sunitinib	Intermediate/poor risk: OS HR (hazard ratio) 0.63; complete response 9%
KEYNOTE-426	Pembrolizumab + axitinib vs sunitinib	Better PFS (progression-free survival) and OS
CheckMate 9ER	Nivolumab + cabozantinib vs sunitinib	PFS 16.6 vs 8.3 months
CLEAR	Pembrolizumab + lenvatinib vs sunitinib	PFS 23.9 vs 9.2 months; response 71%

- **First line:** an IO/TKI (immuno-oncology plus TKI) combination for all IMDC (International Metastatic RCC Database Consortium) risk groups; nivolumab + ipilimumab for intermediate/poor risk only.
- **After IO:** switch to a VEGF TKI; IO re-challenge failed (CONTACT-03, TiNivo-2).

Adjuvant

- **KEYNOTE-564:** pembrolizumab for 1 year after nephrectomy in high-risk clear cell RCC, including resected metastases (M1 NED, no evidence of disease); DFS (disease-free survival) HR 0.68, later OS benefit (HR 0.62). Atezolizumab and nivolumab-based adjuvant trials were negative.

Toxicity: irAEs (immune-related adverse events)

- Colitis, hepatitis, pneumonitis, thyroiditis, hypophysitis, nephritis, myocarditis, rash.

Grade	Action
1	Continue, monitor
2	Hold; prednisolone 0.5–1 mg/kg
3–4	Stop; methylprednisolone 1–2 mg/kg; infliximab or mycophenolate if refractory

- Endocrinopathies: hormone replacement.

Biomarkers

- **PD-L1 does not select patients.** Sarcomatoid tumours respond well. Gene signatures investigational; antibiotics may reduce benefit.

Diagram to draw: T cell–tumour synapse with PD-1/PD-L1 and CTLA-4/B7 blocked.

> **Key points to remember:**
> - IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor only.
> - KEYNOTE-564: adjuvant pembrolizumab, DFS and OS benefit.
> - irAE grade ≥2: hold drug and give steroids.

Q3f. Sperm extraction in Obstructive azoospermia.

Type: SAQ (10 marks) (also answered in 2021-Summer Paper 4, Q2)

Introduction

- **OA (obstructive azoospermia):** no sperm in the centrifuged ejaculate despite **normal spermatogenesis**, from blockage of the ducts.
- Causes: **CBAVD (congenital bilateral absence of the vas deferens)**, vasectomy, epididymal or ejaculatory duct obstruction, hernia-repair injury.
- **Clues:** normal testes, **FSH (follicle-stimulating hormone) ≤7.6 IU/L and testis long axis >4.6 cm**; CBAVD → CFTR (cystic fibrosis transmembrane conductance regulator) testing of both partners.

Indications for retrieval

- CBAVD; unreconstructable obstruction; failed or declined vasectomy reversal; backup at vasoepididymostomy, frozen for IVF (in vitro fertilisation).
- Retrieved sperm need **ICSI (intracytoplasmic sperm injection)**, because sperm from obstructed systems have poor motility.

Techniques

Technique	Method	Merits and limits
MESA (microsurgical epididymal sperm aspiration)	Tubule opened under microscope; aspirated	**Best in OA**: many motile sperm, multiple vials; needs microsurgery
PESA (percutaneous epididymal sperm aspiration)	Fine needle into caput	Simple; fewer sperm, may obstruct epididymis
TESA (testicular sperm aspiration)	23 G needle with suction	Least invasive; immature, immotile sperm
Percutaneous core biopsy	14 G biopsy gun	Fresh retrieval in OA; blind
TESE (testicular sperm extraction)	Open biopsy	When epididymis is absent or empty; gold standard for tissue
Vasal aspiration	Vasotomy, testicular end flushed	At reconstruction

MESA technique

1. Testis delivered; epididymis inspected at 16–25×.
2. Dilated tubule opened with a microknife; fluid checked at once.
3. No motile sperm: close and **move proximally**, up to the efferent ductules (proximal sperm are more motile).
4. Collect by **capillary action** into micropipettes, never suction.
5. **Cryopreserve** in several aliquots; close tubule with 10-0 suture.

Choice and outcome

- Epididymal sperm first; testicular if that fails. Retrieval succeeds in nearly all OA; frozen sperm match fresh (except spinal cord injury).
- **Complications:** haematoma, infection, pain, epididymal obstruction.
- **Also know:** micro-TESE is for NOA (non-obstructive azoospermia).

Diagram to draw: testis and epididymis showing PESA, MESA and TESA sites.

Key points to remember:

- OA: normal FSH and testes; ICSI essential.
- MESA is best: many motile sperm, multiple vials.
- Move proximally; capillary action, never suction.
- TESA/TESE when epididymal sperm absent.

Rapid Revision Sheet – 2024-Winter Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Pathophysiology, classification and management of LUTS. (LAQ)

- ICS: storage, voiding, post-micturition; IPSS 0–7, 8–19, 20–35.
- Static (bulk, capsule) plus dynamic (α1A) obstruction; size ≠ symptoms.
- 5ARI for >40 mL; MTOPS combination cut progression 66%.
- TURP 30–80 mL; HoLEP or simple prostatectomy for >80 mL.

Q2. Etiology, pathogenesis and management of acute renal failure. (LAQ)

- KDIGO: creatinine ≥0.3 mg/dL or ×1.5; urine <0.5 mL/kg/h for 6 h.
- FeNa <1% prerenal, >1% ATN; ultrasound to exclude obstruction.
- Diuretics only for overload; dialysis for refractory AEIOU.
- Post-obstructive diuresis: oral fluids; replace 75% with 0.45% saline if needed.

Q3a. ALPP (SAQ)

- Pves at leak with strain, no detrusor contraction; start at 150 mL.
- <60 ISD; 60–90 equivocal; ≥90 hypermobility.
- VLPP < CLPP; catheter size and volume alter it.

Q3b. Tivozanib (SAQ)

- Selective VEGFR-1–3 TKI; 1.34 mg daily, 3 weeks on, 1 off.
- TIVO-3: PFS 5.6 vs 3.9 months; OS not different.
- FDA 2021: after ≥2 prior systemic therapies.
- TiNivo-2 negative; main toxicity hypertension.

Q3c. 46 XX Male (SAQ)

- 46,XX testicular DSD: SRY on X in about 80–90%; 1 in 20,000.
- Azoospermia always (AZF absent): no sperm retrieval.
- Shorter than Klinefelter; FISH for SRY confirms.

Q3d. Brain stem Death: Biology and clinical implications. (SAQ)

- Raised ICP → Cushing reflex → storm → hypotension, DI, hypothermia, DIC.
- Six reflexes absent; apnoea at $PaCO_2$ ≥60 mmHg.
- Two tests ≥6 h apart; Rule of 100s.

Q3e. Immunotherapy therapy for metastatic RCC (SAQ)

- IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor only.
- KEYNOTE-564: adjuvant pembrolizumab, DFS and OS benefit.
- irAE grade ≥2: hold drug and give steroids.

Q3f. Sperm extraction in Obstructive azoospermia. (SAQ)

- OA: normal FSH and testes; ICSI essential.
- MESA is best: many motile sperm, multiple vials.
- Move proximally; capillary action, never suction.
- TESA/TESE when epididymal sperm absent.